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ནགས་ཚལ་དང་གླིང་ཀ་ཞབས་ཏྲིག་ལས་ཁུངས།

Royal Government of Bhutan
Ministry of Agriculture and Forests
Department of Forests and Park Services



MANAGEMENT AND MARKETING PLAN FOR NON- WOOD FOREST PRODUCTS

(Goney-Wambur Shingmin Tshokpa)



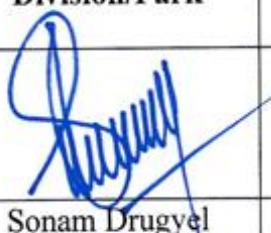
Tsenkhar Gewog
Lhuntse Dzongkhag

(Plan Period: August 2022-July 2027)

APPROVAL SHEET

Name of NWFPs Management and Marketing Group:	Goney-Wambur Shingmin Tshokpa
Dzongkhag:	Lhuntse
Gewog:	Tsenkhar
Village(s):	Goney, Wambur, Lagyel
No. of households:	67
Area (ha)	2401.4
Plan Period:	July 2022 to July 2027

Prepared and submitted for approval

	Chairperson NWFP Management & Marketing Group	Gup of the Gewog	Head, Social Forestry Section of Division/Park	Chief Forestry Officer of Division/Park
Signature:				
Name:	Cheten Dorji	Tashi Penjor	Sonam Drugyel	Karma Tempa
Date:				

REVIEWED AND RECOMMENDED FOR APPROVAL

	Reviewing Officer, SFED	Chief forestry Officer, SFED	Head of the Department
Signature:			
Name:	Sonam Peldon	Karma Jigme Temphe	Lobzang Dorji
Date:			

Approved by

SECRETARY
Ministry of Agriculture & Forests

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PART A: MANAGEMENT AND MARKETING PLAN

1. Introduction

Bhutan is blessed with many NWFPs wherein the local community use it for self-consumption as well as for generating income since the time immemorial. The NWFPs as such plays vital role in improving livelihood of the people residing near and far from the forest. This forest resources could be source of conflict for the people as it is said to be ‘common pool resources’ and it may, at the same time exhaust if not managed carefully. Therefore, Department of Forest and Park Services have been helping the general public in managing resources and meanwhile making people to use the resources sustainably without detrimental effect to conservation.

Goney-Wambur is a chiwog under Tsenkhar Gewog which is located approximately 1300-2000 meter above sea level (masl). The chiwog is connected with feeder road, telephone, electricity and clean drinking water. It is 23 km drive from Gewog Centre, Phawn. The village experience warm summer and cold winter. There are 73 households in the chewog of which 63 households are member in the NWFP management and marketing group. They are mostly agrarian society depending on agricultural activities for their daily livelihood. The farming system is essentially subsistence and main staple food crop grown in the area are rice, maize, wheat etc. They also produce a wide variety of vegetables such as chilies, potatoes, cabbage, radish, beans etc.

The Goney-Wambur NWFP management area falls within Rongmanchu Forest Management Unit. The forest type in the community is both conifer and cool-broadleaved forest. The cool-broadleaved forests dominate the surrounding forest cover. The thick and dense forest coverage provides home to wide range of wild animals such as Wild Boar, Mask Deer, Barking Deer, Bear, Common Leopard, Fox, Dhole, Monkey and variety of birds species.

The non-wood products are distributed sporadically in small pocket of the area. There are total of 13 Non-Wood Forest Products (Table 1) found in the designated area and out of which, four species has economically potential viz. ***Paris pollyphylla*** (Jagpa Thogsumpa), ***Daphne bholua***, ***Rubia cordifolia***, ***Borinda grossa***. Other NWFP products (other than those four products) are also found in the area but can be collected based on the market availability and demand. The potential of NWFP collection may not be in huge quantity, but if harvested and marketed judiciously by the

community people it would help them to earn additional-income. Approximately more than 5934 acres of adjoining forest area provides conducive environment for the community people to collect Non-Wood Forest Products for the Member of **Goney-Wambur Shingmen Tshogpa**.

2. Objectives of Collection and Marketing of NWFP

1. To protect collection of NWFPs by the outsiders.
2. Sustainable harvesting and management of Non-Wood Forest Products.
3. To generate income through sale of Non-Wood Forest Products.

3. Description of NWFPs

The forest of Goney-Wambur has varieties of NWFPs such as *Paris pollyphylla* (Jagpa Thogsampa), *Borinda grossa* (Baa), *Rubia cordifolia* (Tsoe), *Dephne bholua* (Deynap), *Onosma hookeri* (Muhtsi), *Pogostemon amaranthoides* (Namda), *Diplazium sp* (Fern), *Cymbidium spp* (Olachoto), *Acorus calamus* (Shuda), *Elantosteme lineolatum* (Damroo), *Embllica officinalis* (Amla), *Nephrolepsis cordifolia* (Khowchi khiwchi), *Viscum articulatum* (Ngashing Jorma), *Otochilus lancilabius* (Olatse). The most of the NWFPs are found in upper ridge of Wambur and Lagyel village which is dominated by broadleaved forest while Amla and Khowchi is found in lower elevation especially in Chir-pine dominated area. The *Borinda grossa* and *Daphne bholua* are found abundantly in Lagyel area and local people extract *Borinda grossa* for both domestic and commercial purposes. The *Daphne* is also one of the product that has economical prospect in the area although people have not yet collected. . The group mainly focuses on collection of *Borinda grossa*, *Rubia cordifolia*, *Daphne bholua* and *Paris polyphylla* as the main products for commercial marketing and *Daphne bholua* will be collected only if there is demand from the buyers/traders. NWFPs such as ferns, mushroom, etc will be collected mainly for domestic purposes and for commercial if there is demand.

Table 1: Brief description of NWFP for commercial harvesting

NWFP species	Harvesting area(s) name	Part harvested	Use purpose (traditional & others)	Harvesting season(months)	Estimated quantity that can be collected in a year
<i>Borinda grossa</i>	Rewapang Juug, Sadhu Brangsa, Rashu Goenpa	Culms	Weaving bamboo mates/handicraft	December to February	1000 culms
<i>Rubia cordifolia</i>	Lagay, Wambur, Kiphu, Goney	Upper part of plant	red dye and medicinal	November to February	1500 kg
<i>Paris polyphylla</i>	Damtshang, Rewa, Phuningla Duptsi Gonpa, Shogtshangla	Rhizome	Curing burns, cuts, diarrhea, dysentery, fever, stomachache	October to November	60 kg
<i>Daphne bholua</i>	Lagay, Tsanchotla, Goney	Bark	Traditional paper	March to June	1400 kg

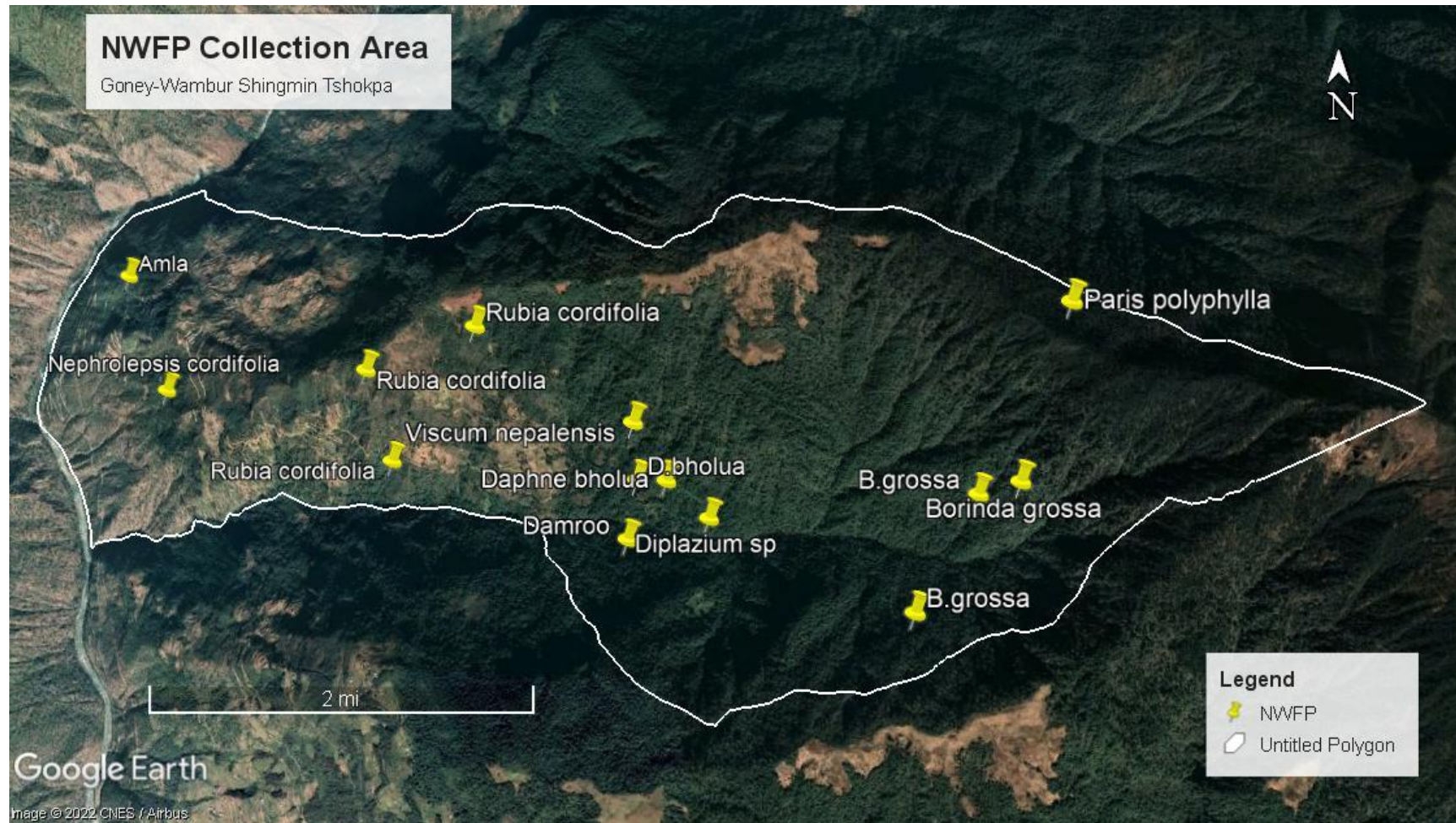
Table 2: Harvesting guidelines and description for other NWFP

Sl. No	Name		Parts Used	Purpose	Time of Harvesting (month)	Harvesting Guideline
	Scientific	Local				
1	<i>Onoma hookeri</i>	Muhtsi	Roots	Religious purpose	September to December	Dig up only matured rhizomes. Leave the mother plant intact with small part of root in the soil. Harvest rhizome during harvesting season. Do not uproot entire plant.

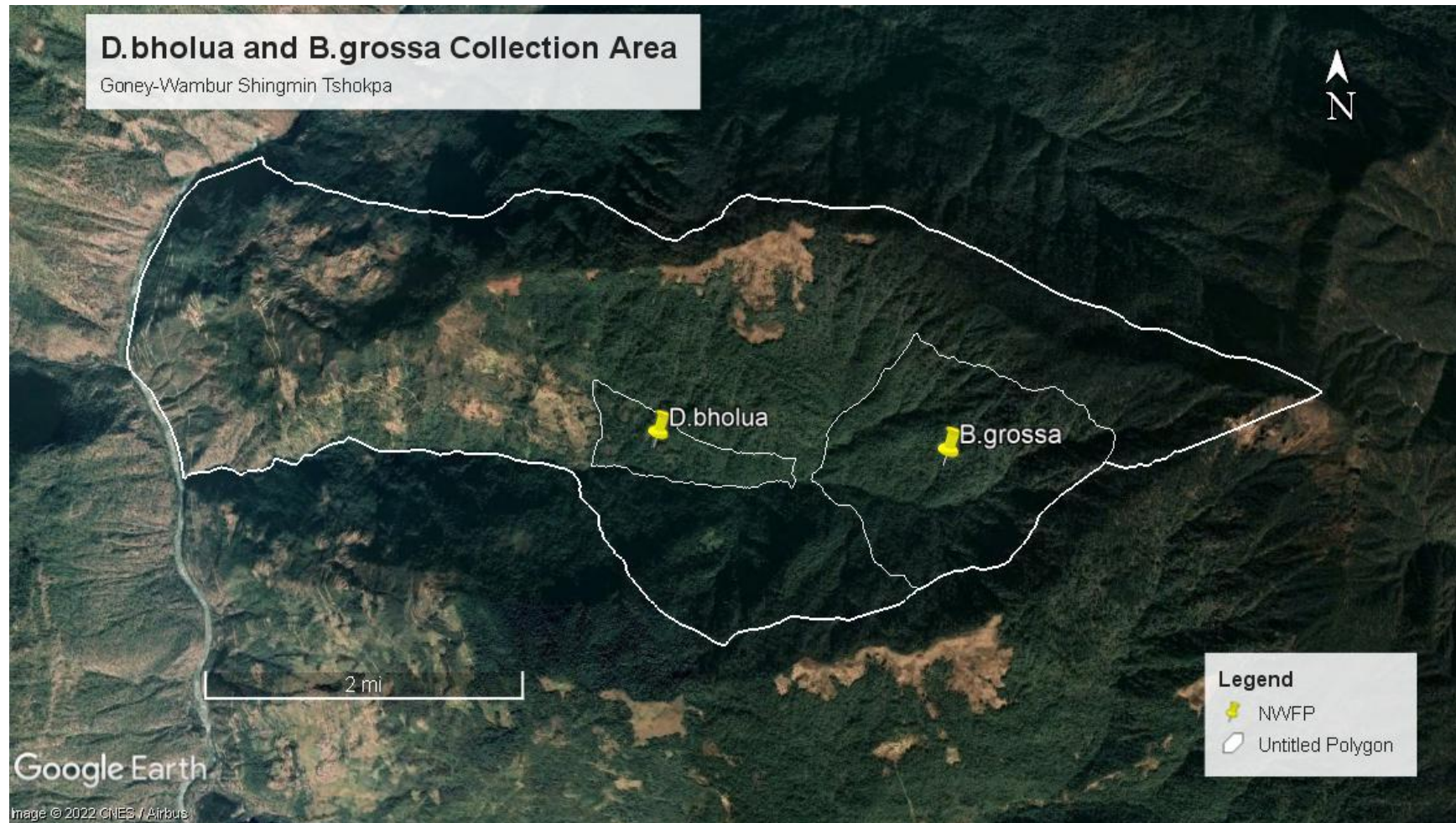
2	<i>Pogostemon amaranthoides</i>	Namda	Leaves	Vegetable	May to September	Cut tender shoot only. Retain 1% of mother plants in clump
3	<i>Diplazium spp</i>	Fern	Shoot	Vegetable	June to September	Cut tender shoot only. Retain 1% of mother plants in clump
4	<i>Cymbidium spp</i>	Olachoto	Tender parts	Vegetables	October to January	Cut tender shoot only. Retain mother plants in clump
5	<i>Acorus calamus</i>	Shuda	Roots	Medicine	Whole year	The plant is uprooted either by hand or using hoes, cut the root parts and plant back the aerial stem
6	<i>Elantosteme lineolatum</i>	Damroo	Leaves	Medicine and food	9 th to 12 th Bhutanese month	Cut tender shoot only
7	<i>Emblica officialnalis</i>	Amla	Leaves	Medicine/food	January to March	Plucking by hand or shaking with small stick Picking fruits from the forest floor. Don't cut down tree in order for fruits to collect.
8	<i>Nephrolepis cordifolia</i>	Khowchi khiwchi	Root	Medicine	Whole year	The plants are uprooted either by hand or hoe. Retain few shoots as mother trees
9	<i>Viscum articulatum</i>	Ngashing Jorma	Shoot		March to May	The plants has to be cut 20cm above ground (retain few plants as seed trees)

4. Maps of NWFP Collection Area

4.1 Map of the NWFP Harvesting Areas

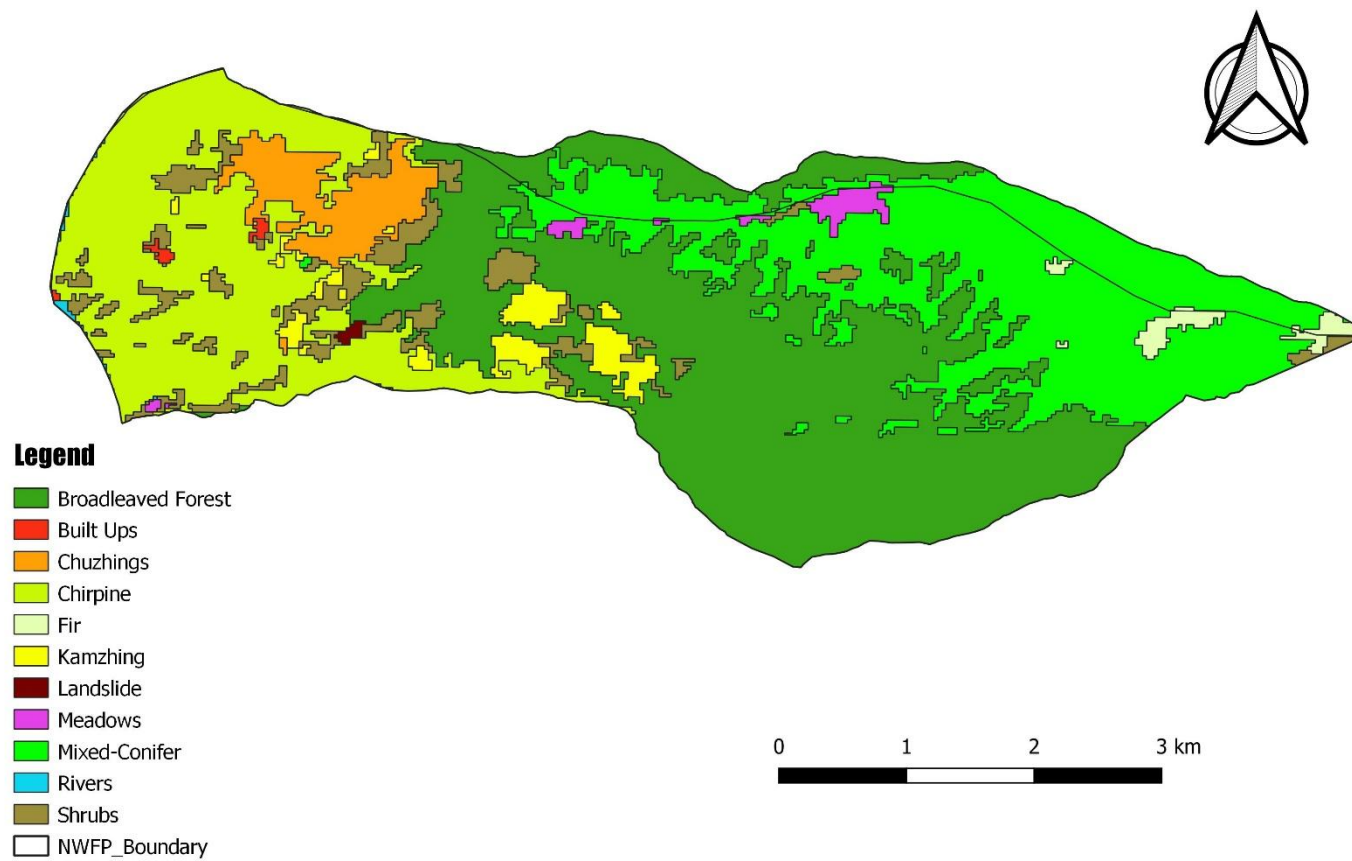


4.2 Map of *Borinda grossa* and *Daphne bholua* Extraction



4.3 Land Use Map of NWFP Area

Landuse and Landcover Map of NWFP Collection Area



5. Sustainable NWFP Harvesting Prescriptions

5.1 Domestic purpose

Any individuals harvesting NWFPs for domestic consumption should inform the Chairperson before proceeding for collection.

Harvesting of NWFPs for domestic purposes can be done any time with the realization of Royalty. Meanwhile, members shall comply with the restrictions (Reedam/Laadam) imposed by the village.

Maximum harvesting limit of *Borinda grossa* for domestic purposes is 200 numbers per household/season.

NWFPs like mushroom, damroo and fern for domestic consumption can be harvested anytime as per the requirement of the individual members.

5.2 Commercial purpose

Commercial harvesting should be done together involving all the members after obtaining collection permit from nearby forest office.

Any members willing to harvest the *Paris polyphylla*, *Rubia cordifolia* and *Borinda grossa* for commercial/domestic purpose, should inform the Chairperson before collection so that the chairperson could process for obtaining the collection permit.

The commercial extraction of NWFPs shall be done only by the members; the outsiders are strictly restricted for commercial extraction.

Table 3: Harvesting prescription for sustainable harvesting of NWFP

NWFP species	Harvesting prescription for sustainable harvesting
<i>Borinda grossa</i>	Cut culm at the base or at first long internodes above the nodes. Leave bamboo shoot or culm of less than one year old intact. Do not cut culm when new shoots are sprouting. Harvest culm in condensed culm in U shape
<i>Rubia cordifolia</i>	Use sharp scissors for the harvesting of the upper parts of the plants. The plant has to be cut from 5cm above the ground. Do not uproot entire plants.

	Leave at least 10% of the plants without collecting to ensure natural regeneration.
<i>Paris polyphylla</i>	Dig up only mature rhizomes. Leave the mother plant intact with small part of root in the soil. Harvest rhizome on during harvesting season. Do not uproot entire plant.
<i>Daphne bholua</i>	Cut the plant 10cm above ground level. Cut stem that are more 3cm in diameter. Do not damage the remaining young stands/seedlings or saplings

6. Marketing and Sales Plan

- The NWFPs will be sold to the buyers/traders by the group at least involving one fourth of the member.
- No individual shall be permitted to sell without the consent of the group members.
- Standard/Harvestable size (2-4 cm) and length (5 m) of *Borinda grossa* for commercial purposes have to maintain. One back load should consist of 30-40 nos. for men and 15-20 nos. for women and children. It shall be sold at a rate of Nu.6/- which includes the royalty for outsiders and Nu.2 for group members.
- One kg of dried *Daphne bholua* shall be sold at Nu.50/- including the royalty irrespective of quality.
- Cost of *Paris polyphylla* has been fixed at the rate of Nu.2500/ kg including the royalty.
- The dried *Rubia* shall be sold at the rate of Nu.50/- per kg and shall not sell the fresh *Rubia*.
- Permit fee and royalty of NWFPs should be paid as per the royalty of the DoFPPs.
- The other NWFPs listed in the annexure shall be sold as per the commercial rate fixed by the DoFPPs.
- All the NWFPs shall be sold to the trade license holder only.
- The amount received or earned from the sale of NWFPs will be deposited in the group account.

The marketing and sales plan for different types of NWFPs that are to be collected by the group, its potential markets where the group shall sell products with approximate quantity, price per unit and the total value of the expected sales are described in the table below:

Table 4: Marketing and financial projections

<i>NWFP</i>	<i>Potential market</i>	<i>Estimated quantity for sale per season (unit)</i>	<i>Expected price per unit</i>	<i>Total (Nu)</i>
<i>Borinda grossa</i>	Local Market	10000 culms	6/-	60,000
<i>Rubia cordifolia</i>	Within Bhutan/India	7500 kg	50/-	3,75,000
<i>Paris polyphylla</i>	Within Bhutan	300 kg	2500/-	7,50,000
<i>Daphne bholua</i>	Local market	7000 kg	50/-	3,50,000
Total Amount				1,535,000

Note: The marketing and sales plans for NWFPs list in the above table (qty) is estimated for 5 years.

7. Marketing Arrangements

The Chairperson shall collect demand of NWFPs before onset of any product collection season with support of concerned Forestry Officials. Then, the marketing of NWFPs will be organized and arranged as per the procedures given below:

- Marketing of products will be always group based. However, the Chairperson will arrange selling point either at Lhuntse, Mongar or in nearby villages and towns. The members will collect forest/wild vegetable items from different blocks on rotational basis. It will be taken to vegetable market along with other agriculture products. OR
- Members are not allowed to accept any advance payment from buyers at any circumstances as such practices deteriorate quality of products and also encourage over exploitation of resources.

8. Resource Assessment

The resource assessment of two potential NWFP viz. *Borinda grossa* and *Daphne bholua* in the area has been carried out to estimate the amount that can be harvested on sustainable basis from the area.

8.1 *Borinda grossa*

The culms of *Borinda grossa* is long, straight and it is useful for weaving durable fencing mats and roofing mats, baskets, containers, hats.

8.1.1 *Field data collection*

Borinda grossa resource assessment was carried at Lagyel area following the stratified sampling method in which area is divided into different blocks according to the location and density. The plant is measured randomly in the chosen sample plots. The quantitative assessment was carried out by laying the sample plots of 10 m x 25 m along the slope direction. From each sample plot, the bamboo stock is measured including the condition of the clump, number 1+ year, 2+ year and 3+ year culms and the number of culms that the local people consider harvestable. These harvestable culms are mostly 1+ year, 2+ years are in good health and are straight, have length of five m and a diameter of 2 to 4 cm. Resource assessment was done by collecting the data about the locations and quantities of species present in the area. Based on the results of this survey, the area is divided in to 3 blocks according to the availability of the resources on the Google map. Out of 3 blocks, the farmers has decided and agreed to collect the *Borinda grossa* from the block alternatively. The members also decided to collect *Borinda grossa* based on AHL and as per the guidelines.

8.1.2 *Data analysis*

Data collected from the sampling plots were entered in the excel sheet and then sorted cleaned for further analysis work.

8.1.3 *Calculation of Annual Harvesting Limits (AHL)*

AHL: 0% of the shoots +33% of the total stock of 1+ year and 2 + years old culms+70% of the 3+ years old culms (Table 5).

Based on the results of the inventories, the total harvestable culms were calculated for each sub block. The AHL for 1+ year and 2+ years old culms from the clump was set at 33% of total culms. This will ensure the regeneration of the species because the shoots of less than 1 year old culms will be retained and provide the 1+year culms for following year. There will be also enough 1 + year culms to provide 2 + years old culms for next season. Meanwhile, 70% of 3 +

years culms can be harvested and some will be left in the area to provide physical support to the younger culms. Based on the above AHL calculations, people decided to follow the following conditions;

1. Maximum of 33% of 1+ year and 2+ year old culms per clumps will be harvested and 67% will be left intact in the area for regeneration.
2. 70% of 3+ years old culms will be harvested and rest will be left in the area to provide physical support to the younger culms.

Table 5: Data analysis of *B. grossa*

Resource assessment compilation form for Borinda grossa						
Block name:	Lagay Bamboo	Dates of bamboo	26-28/01/2022			
Sub-block (Number)	Name of the sub-block	Area (ha)	% of productive	Productive area (ha)	Number of sample	Remarks
1	Rewapang Juug	94.5	34.4	32.5	5	
2	Sadug Brangsa	108	34.6	37.4	6	
3	Rashu Goenpa	83.3	27.1	22.6	6	
Age class	Number of culms	Sub-block 1	Sub-block 2	Sub-block 3	Total in Block	
1 + year old bamboo culms	Total counted in sample plots	362	729	298	1389	
	Total harvestible in sample plots	218	460	208	886	
	Total in sub-block*	94120	181764	44899	320783	
	Total harvestible in sub-block*	56680	114693	31339	202712	
2 + years old bamboo culms	Total counted in sample plots	405	743	285	1433	
	Total harvestible in sample plots	294	492	179	965	
	Total in sub-block*	105300	185255	42940	333495	
	Total harvestible in sub-block*	76440	122672	26969	226081	
3 + years old bamboo culms	Total counted in sample plots	591	791	335	1717	
	Total harvestible in sample plots	111	266	119	496	
	Total in sub-block*	153660	197223	50473	401356	
	Total harvestible in sub-block*	28860	66323	17929	113112	
Total Stock in Sub-Block		353080	564241	138312	1055633	
Total Harvestible in Sub-Block		161980	303688	76237	541905	
AHL-33% (1 & 2 + years)		136360	245237	57757	439353	
AHL-70% (3+ years)		107562	138056	35331	280949	

8.2 *Daphne bholua*

Daphne bholua (Deynap) is a shrub of 1-3.5 m in height and occurs in evergreen oak, spruce, hemlock, fir and blue pine forests at an altitude of 1950-3400 m. It is used for the production of traditional paper. *D. bholua* is the only *Daphne* species that is also used as an ornamental plant because of its strong fragrant flowers.

8.2.1 Field data collection

The resource assessment for *D.bholua* is carried out at Goney and Lagyel area following a systematic sampling design with the sampling plots of 10 m x10 m. In each sampling plot only the diameter of the stem was recorded. Research has shown that there is a direct correlation between the diameter and the total dry bark yield, thus more data are not needed. The diameter of Deynap and Deykap were measured at 10 cm from the ground level, referred to as D10. The diameter of all stems more than 1 cm at D10 were recorded. Plants with a smaller diameter were recorded as < 1 cm. The diameters of multi-stemmed plants were converted into a single stem diameter. Based on this single stem diameter, the total dry bark weight were estimated based on a formula. Dead plants are not measured. A plant is declared dead, when it has no green leaves during vegetation time. If it is estimated that the plant has lost more than half of its bark, it is also not measured; if less than half of the bark is lost, the plant is measured as usual.

8.2.2 Data analysis

Based on research carried out by C. Hadorn and P. Wangda (1999), a formula was developed to estimate the total dry bark yield of Deyknap, *Daphne bholua* and *Daphne sureil* (there are no data available about the other *Daphne* species) for whole of Bhutan:

$$\mathbf{Wd = 0.02559 * D10^{2.27221}}$$

In which Wd is the estimated total dry bark weight in g, and

D10 is the diameter at 10 cm from the ground in mm

Based on this formula the total dry bark weight were estimated as a function of the diameter at 10 cm from the ground. According to the same research of C. Hadorn and P. Wangda (1999), some 47% of the total dry bark weight cannot be used and is considered as waste bark (for all diameters),

the remaining 53% is the usable bark. As per the calculations the total dry bark weight which can be collected is 7121644.98 g (7121.6 kg).

Table 6: Data analysis for *D. bholua*

Data analysis per block							
Block name:		Daphne Extraction Block					
Number of plants with a diameter of less than 1 cm:							
Sub-block (name)	Area (ha)	No of sample plots	Total no of plants of D10<1cm	Average no of plants of D10<1cm per sample plot	Average no of plants of D10 < 1 cm per ha	Total no of plants of D10 < 1 cm in sub-block	
	(a)	(b)	(c)	(=c/b)	(=100*d)	(=a*e)	
				(d)	(e)	(f)	
D-Lagyel	75.8	70	539	7.7	770	58366	
Number of plants with a diameter of 1-1.9 cm:							
Sub-block (name)	Area (ha)	No of sample plots	Total no of plants of D10: 1-1.9 cm	Average no of plants of D10: 1-1.9 cm per sample plot	Average no of plants of D10: 1-1.9 cm per ha	Total no of plants of D10: 1-1.9 cm in sub-block	
	(a)	(b)	(c)	(=c/b)	(=100*d)	(=a*e)	
				(d)	(e)	(f)	
D-Lagyel	75.8	70	206	3	294	22307	
Number of plants with a diameter of 2-2.9 cm:							
Sub-block (name)	Area (ha)	No of sample plots	Total no of plants of D10:2-2.9 cm	Average no of plants of D10: 2-2.9 cm per sample plot	Average no of plants of D10: 2-2.9 cm per ha	Total no of plants of D10: 2-2.9 cm in sub-block	
	(a)	(b)	(c)	(=c/b)	(=100*d)	(=a*e)	
				(d)	(e)	(f)	
D-Lagyel	75.8	70	319	5	456	34543	
Available usable dry bark weight per sub-block of plants with diameter of > 3.0 cm:							
Sub-block (name)	Area (ha)	No of sample plots	Total Wd of plants with diameter of > 3 cm	Average Wd per sample plot	Average Wd per ha	Total Wd in sub-block	Total usable bark
	(a)	(b)	(c)	(=c/b)	(=100*d)	(=a*e)	(53% off)
				(d)	(e)	(f)	
D-Lagyel	75.8	70	124089	1772.7	177270	13437066	7121644.98
Note: Wd is the weight of bark in gram(a)							

Note: Wd is the weight of bark in gram(g)

PART B: By-laws: Group Regulations and Norms

1. Members of the NWFP management & marketing group	Village	Number of member households	Gewog
	Goney	3	Tsenkhar
	Lagay	7	
	Wambur	50	
	Total	63	
	Responsibilities and rights of members:		
	• All the members shall be responsible for sustainable management of NWFPs. • The NWFPs shall be harvested during suitable season as prescribed in management plan. • The NWFPs group members shall follow and respect the instruction of executive members. • All members shall be vigilance for illegal harvesting of NWFPs by non-members and report to forest office/executive members. • All the group members have right to harvest NWFPs and market as per this plan. • The members will collect NWFPs as per the directives of the Chairperson based on demand of the trader/buyers but not exceeding the annual harvesting limit. • Members should inform the Chairperson/Forestry Official of any person extracting rural timber from NWFP harvesting area so as to stop the illegal harvesting of the NWFPs.		

2. Executive committee	
2.1 Chairperson	Name of Chairperson: Cheten Dorji
	Responsibilities of Coordinator/ Chairperson: • The Chairperson has power to impose the fines and penalties if any members do not follow his instruction (meeting, timing). • Chairperson shall obtain the NWFPs collection permit from forest office during the collection season. • Inform or announce the date of NWFPs collection to each member. • Should coordinate the NWFPs harvesting. • Shall contact with the buyer/traders on the required quality of the NWFPs and inform the members accordingly. • Shall look for the buyers/traders. • To chair/coordinate all meetings related to this plan. • Resolve disputes and conflicts among members. • Take care of overall financial arrangements. • Attend/participate trainings/workshop/study tour when called.
2.2 Secretary	Name of Secretary: Jambay Dorji
	Responsibilities of Secretary: • Maintain/Keep the sale records. • Maintain the minutes of meeting. • Attain/participate trainings/workshop/study tour when called. • Assist the Chairperson. • Keep detail record of NWFPs harvest and submit to Chairperson
2.3 Treasurer	Name of Treasurer: Kinzang Tshomo
	Responsibilities of Treasurer:

	• Maintain cash transaction records. • Collect membership fees, incentives/compensation fees and deposit in bank • Look overall financial transaction in collaboration with the Chairperson and Secretary. • Officiate Secretary. • Keep detail record of NWFPs sale data. • Present account statement to the group during meeting • Issue money receipt after collection of fines and penalties
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2.4 Resungpa	Name of Resuups: Geley Wangchuk, Nidup Wangdi, Sonamla, Tenzin Choiphel, Thinley Wangchuk
	Responsibilities of Resuups: • Patrol the NWFP area and report to Chairperson.

2.5 Compensation to the executive committee for marketing	The group members discussed and decided to compensate the executive members with incentives of Nu.250 whenever there is sale of NWFPs. The incentives will be paid from the amount received from the sale of NWFPs which are extracted only from their Jurisdiction. If the group buy products from other areas, the compensation shall not be paid to any executive members.
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3. Membership fee and other conditions	• Each member shall contribute Nu.150/- once in a year as a membership fee and shall be collected during the annual meeting. • Any new member shall pay 5% of the total amount collected till date • The fund will be deposited in the bank in the name of group counter signed by coordinator, secretary and treasurer. • Anyone wishing to withdraw from the group will have to put application stating all the reasons and no objection in future. The decision will be made during meeting to leave the group. • In unfavorable situation if one member wish to resign he/she will be allowed with equal benefit sharing. • Once any member who had resigned from the group again wishing to join group will not be entertained. Any collection NWFP within the designated area is prohibited for them.
4. Community development fund and utilization of fund	The group decided not to have community development fund separately. The amount received or earned from the sale of NWFPs will be fully deposited in the group account which is already maintained. In any community based developmental activities or for spiritual matters, the group members will discuss and make the fund arrangements thereof. During such critical instances, the discussion shall be carried out with not less than 50% of members.

5. Conflict resolution	• Conflicts or disputes within the NWFP Management Group, which cannot be resolved by the group members shall be forwarded to the Chairperson for settlement. • If the case cannot be resolved, it shall be forwarded to the Gewog Administration via Chiwog Tshogpa. • In case there is a conflict between the NWFP Group and outsiders, the case shall be forwarded to the Divisional Forest Office and Gewog Administration by the Chairperson.
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6. Penalties	• If any member sells the NWFPs directly to the traders without the notice of the group executive committee members shall be penalized as per the FNCRR, 2017. • If the non-members are found doing illegal/taking of NWFPs shall be dealt as per the FNCRR, 2017. • The group members shall sell the NWFPs to the valid trade license holder only, if not the group shall be penalized as per the FNCRR, 2017. • The annual meeting shall be conducted on 30th month of December. Any members remaining absent in meeting should pay Nu.500/-as per the local village fine rules. • If the committee members fail to attend the meeting, he/she shall be levied a fine of Nu.700. The collected amount shall be deposited in the group saving account. However, during the times of death and sickness consideration will be made. • If any executive member's misuses the group fund, he/she has to pay double the amount within the date line given by the group.
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7. Benefit sharing	• The group fund will be shared equally among the members • The group members are not allowed to collect NWFPs for individual marketing • The collection of NWFPs by outsider is strictly prohibited • Only small quantity is allowed to collect for self-consumption
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8. Meetings	The group members will have one time meeting in a year which will happen on last week of December month. The group decided to conduct meeting at Wambur in Treasurer's residence as it falls in center of the NWFP collection area. Should there be any ad-hoc meetings, the Chairman will inform the group and conduct at any time.
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9. Authority of the Department	As per Chapter III, section 155 of FNCRR, 2017, the Department shall: • Suo moto or on the request by the Non-Wood Forest Product Management Group (NWFPMG) take action against violator of the Act, Rules or Management Plan; • Inspect, monitor, review the records of and otherwise supervise the activity of the NWFPMMG; and • Take action including cancellation of the NWFMGM certificate if the NWFMGM or any of its members who violate the provision of the Management Plan or any provision under this Chapter
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10.Cancellation of NWFP Management Certificate	As per Chapter III, section 156, 157 & 158 of FNCRR, 2017: • The certificate shall be cancelled, if the performance of NWFMG is not as per prescriptions of the approved management plan or the sustainability of the resource is not ensured • In the event the members of the NWFMG decide to cancel NWFP Management certificate in accordance with this Rules, the NWFP Management Certificate shall be deemed cancelled on the date of such decision being notified to CFO in writing and the NWFMG shall surrender the Certificate to the Department • The CFO shall have the right to enquire about the reason for such cancellation to ensure that no offence has been committed and shall notify the NWFMG on the cancellation of the certificate.
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Annexures I: NWFP Member List

No	Name	Village	Signature	Signature
1	Sonam Rinchen	Lagay		
2	Tshenwang Peldon	"		
3	Cheten Dorji	"		
4	Tendrel Zangmo	"		
5	Rinzin Tshomo	"		
6	Galey Tshering	"		
7	Kezang Tshering	Gongid		
8	Kuenzang Rangdrol	"		
9	Nidup Wangdi	"		
10	Sangay Wangdi	Trenchola		
11	Namgay Wangdi	Lagay		
12	Sonam Beki	Kiphu		
13	Toumo	"		
14	Yangdon	Dagorthang		
15	Beki Mo	"		
16	Tandin Wangmo	"		
17	Cheki Wangmo	"		
18	Pema Yangzom	Khimgong		
19	Sherab Pelmo	"		
20	Beki Yangzom	"		
21	Choning Harbu	"		
22	Beki Pelmo	"		

	Yuden Mo	Khangong	W	W
	Phuntsha Wangmo	"	sgu	sgu
4	Choki Wangmo	Khablang		
25	Jamboy Dorji	"		
26	Pema Lhamo	"		
27	Tshomo	"		
28	Norbu	"		
29	Karma Cheki	"		
30	Rinzin Mo	"		
31	Yeshe Mo	"		
32	Tanala	"		
33	Kinlay Wangmo	"		
34	Pema Choden			
35	Tshering Dorji	"		
36	Ugyen Wangmo	"		
37	Ugyen Wangmo	Yanglong		
38	Tsieten Lhamo	"	But	But
39	Rinchen Dema	"		
40	Dorji Wangmo	"		
41	Sonam Dema	Bara		
42	Bumpa Mo	Gurtong		
43	Shacha	Khablang		
44	Tshering Yangki	"		

	Jigme Thinley	Gajenshang	TH	TH
	Pema Tshoki	Gedong		
47	Hangmo	Jampang		
48	Pema Gyaltshen	"	TH	TH
49	Karma Tenzin	Sokorum		
50	Tashi Chopel	Sokchen		
51	Tshewang Penjor	"		
52	Lhacham Choden	Gang	LB	LB
53	Choden	"	TH	TH
54	Kinzang Tshomo	Jarkang		
55	Karma Chopel	"		
56	Sangay Gyeltshen	Chungka		
57	Thinley Hangchuk	"	TH	TH
58	Chimi Dorji	"	TH	TH
59	Sonam Chojay	"		
60	Jigme Singye	"		
61	Ugyen Dema	Bodma	TH	TH
62	Tashi Hangmo	Dowajigbala		
63	Dorji Dema	Gedong		

Annexure II: Block description for *Borinda grossa*

Block description form for <i>Borinda grossa</i>					
Date	26-28/01/2022				
Village	Lagyl	Name of the bamboo area	Rawapang Juug, Sadug Brangsa & Rashu Goenpa		
Block Name	Lagay Bamboo Production Area	Total area (ha)	285.8		
Access	What is the walking distance from the nearest village to the block				
	less than 3 hours	Within a day	More than a day		
		√			
	What is the walking distance from the block to the nearest road				
	less than 3 hours	Within a day	More than a day		
Slope		√			
	Very steep >100%	Steep 70-100%	Moderate 45-70	Gentle <45%	
			√		
Forest description	Dominant and associate species, canopy cover: Quercuss sp, Rhododendronn sp				
Sub-Block	No	Density	Area (ha)	Productive area (%)	No. of sample plots
	1	Medium	94.5	34.4	5
	2	High	108	34.6	6
	3	Low	83.3	27.1	6
Bamboo flowering	Sporadic	Gregarious	Last time of the flowering	None	
Planation potential		√	2007-2008		
	Sub-blocks	Has plantation potential	No potential	Not applicable	
	1		√		
	2		√		
Current management	3		√		
	Current management practices in the entire block:				
	Threat	Description			
	Fire	No threat			
	Grazing	Mild			
Current threats	Illegal exploitation	Very less chance			
	Over exploitation	Less chance			
	Other				
	Opportunity	Future opportunity for the entire block: Can be used as mass bamboo production area and commercialization beyond locality			

Annexure III: Block description for *Daphne bholua*

Appendix 2: Block description form for Deynap and Deykap

Date: 4-14/2/2022

Village: Lagayel Name of Deynap/ Deykap area: Deynap.

Block name: D-Lagayel Total area (ha): 75

Access	What is the walking distance from the nearest village to the block?		
	Less than 3 hours ☒	Within a day	More than a day
	What is the walking distance from the block to the nearest road?		
	Less than 3 hours ☒	Within a day	More than a day

Slope	Very steep	Steep	Moderate	Gentle
	>100%	70-100%	45-70%	< 45 % ☒

Forest description	Dominant and associate species, canopy cover: Symplocus sp, Schima sp.
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Sub-block	No.	Density	Area (ha)	No of sample plots
	1		75	70
	2			
	3			

Plantation potential	Sub-blocks	Has plantation potential	No potential	Not applicable
	1	☒		
	2			
	3			

Current management	Current management practices in entire block: No management.
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Current threats	Threat	Description
	Fire	
	Grazing	☒
	Illegal exploitation	
	Over-exploitation	☒
Other:		

Opportunities	Future opportunities for the entire block: Plantation. and
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Annexure IV: *B. grossa* data analysis

Resource assessment compilation form for Borinda grossa						
Block name:	Lagay Bamboo Production Area	Dates of bamboo resource assessment:		26-28/01/2022		
Sub-block (Number)	Name of the sub-block	Area (ha)	% of productive area	Productive area (ha)	Number of sample plots	Remarks
1	Rewapang Juug	94.5	34.4	32.5	5	
2	Sadug Brangsa	108	34.6	37.4	6	
3	Rashu Goenpa	83.3	27.1	22.6	6	
Age class	Number of culms		Sub-block 1	Sub-block 2	Sub-block 3	Total in Block
1 + year old bamboo culms	Total counted in sample plots		362	729	298	1389
	Total harvestible in sample plots		218	460	208	886
	Total in sub-block*		94120	181764	44899	320783
	Total harvestible in sub-block*		56680	114693	31339	202712
2 + years old bamboo culms	Total counted in sample plots		405	743	285	1433
	Total harvestible in sample plots		294	492	179	965
	Total in sub-block*		105300	185255	42940	333495
	Total harvestible in sub-block*		76440	122672	26969	226081
3 + years old bamboo culms	Total counted in sample plots		591	791	335	1717
	Total harvestible in sample plots		111	266	119	496
	Total in sub-block*		153660	197223	50473	401356
	Total harvestible in sub-block*		28860	66323	17929	113112
Total Stock in Sub-Block			353080	564241	138312	1055633
Total Harvestible in Sub-Block			161980	303688	76237	541905
AHL-33% (1 & 2 + years)			136360	245237	57757	439353
AHL-70% (3+ years)			107562	138056	35331	280949

Annexure V: *D.bholua* data analysis

Data analysis per block							
Block name:		Daphne Extraction Block					
Number of plants with a diameter of less than 1 cm:							
Sub-block (name)	Area (ha)	No of sample plots	Total no of plants of D10<1cm	Average no of plants of D10<1cm per sample plot	Average no of plants of D10 < 1 cm per ha	Total no of plants of D10 < 1 cm in sub-block	
	(a)	(b)	(c)	(=c/b)	(=100*d)	(=a*e)	
				(d)	(e)	(f)	
D-Lagyel	75.8	70	539	7.7	770	58366	
Number of plants with a diameter of 1-1.9 cm:							
Sub-block (name)	Area (ha)	No of sample plots	Total no of plants of D10: 1-1.9 cm	Average no of plants of D10: 1-1.9 cm per sample plot	Average no of plants of D10: 1-1.9 cm per ha	Total no of plants of D10: 1-1.9 cm in sub-block	
	(a)	(b)	(c)	(=c/b)	(=100*d)	(=a*e)	
				(d)	(e)	(f)	
D-Lagyel	75.8	70	206	3	294	22307	
Number of plants with a diameter of 2-2.9 cm:							
Sub-block (name)	Area (ha)	No of sample plots	Total no of plants of D10:2-2.9 cm	Average no of plants of D10: 2-2.9 cm per sample plot	Average no of plants of D10: 2-2.9 cm per ha	Total no of plants of D10: 2-2.9 cm in sub-block	
	(a)	(b)	(c)	(=c/b)	(=100*d)	(=a*e)	
				(d)	(e)	(f)	
D-Lagyel	75.8	70	319	5	456	34543	
Available usable dry bark weight per sub-block of plants with diameter of > 3.0 cm:							
Sub-block (name)	Area (ha)	No of sample plots	Total Wd of plants with diameter of > 3 cm	Average Wd per sample plot	Average Wd per ha	Total Wd in sub-block	Total usable bark
	(a)	(b)	(c)	(=c/b)	(=100*d)	(=a*e)	(53% off)
				(d)	(e)	(f)	
D-Lagyel	75.8	70	124089	1772.7	177270	13437066	7121644.98
Note: Wd is the weight of bark in gram(g)							

Note: Wd is the weight of bark in gram(g)

Annexure VI: Data Collection Pictures

